

An Introduction to Post-Quantum Cryptography

Manipal School of Information Sciences
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Friday, 24 April 2026

- ✓ Classical Cryptography
 - Threats from Quantum Computing
 - Mitigating the Quantum Threats
- ✓ Post-Quantum Cryptography (PQC)

✓ *involves hands-on Python programming*

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|----------------------------------|--------------------|
| ✓ Classical Cryptography | ~2 hours |
| – Threats from Quantum Computing | ~20 minutes |
| – Mitigating the Quantum Threats | ~20 minutes |
| ✓ Post-Quantum Cryptography | ~1.5 hours |

✓ *involves hands-on Python programming*

- Classical Cryptography **~2 hours**
 - Primitives
 - Protocols
 - Computational Hardness Assumptions
- Threats from Quantum Computing **~20 minutes**
- Mitigating the Quantum Threats **~20 minutes**
- Post-Quantum Cryptography **~1.5 hours**
 - Quantum-Resistant Computational Problems
 - Principles of Lattice-Based Cryptography
 - ML-KEM and ML-DSA schemes

Cryptography is fun because it is a blend of

- Mathematics
- Computational Complexity Theory
- Algorithms and Programming
- Systems Security

Science and engineering in equal parts

Mathematics

Computational Complexity Theory

Algorithms and Programming

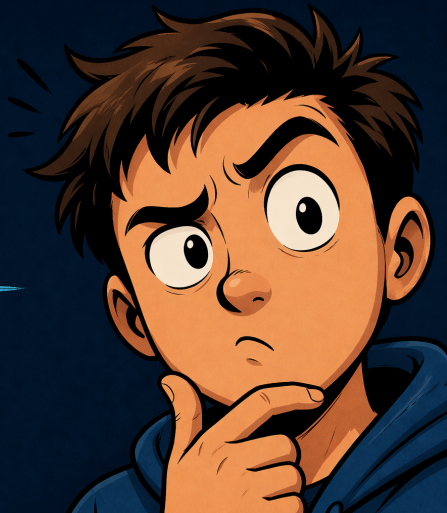
Systems Security

Do not hesitate to ask questions!

But Wait...

Introduction

**WHY
SHOULD
I CARE?**



Cryptography is pervasive

- Aadhaar
- UPI
- TLS (*and* HTTPS)
- Signal/WhatsApp
- GitHub (*with* SSH authentication)
- WireGuard (VPN support in Linux kernel)
- Cryptocurrencies
- ...

It is believed that the *quantum devices* of the future have the computational power to break cryptography schemes based on asymmetric/public key methods.

Today, we will learn a few ideas that are meant to protect *classical* cryptography from *quantum adversaries*.

Classical Computers and Algorithms

Quantum Computers and Quantum Algorithms

Post-Quantum Cryptography (PQC)

- It is **NOT** post-quantum computing
- It is *simply* post-quantum cryptography
- You *don't* need a quantum computer to do PQC!

PQC Readiness In the Works...

NITI Aayog's Roadmap.

[niti_roadmap_quantum_powered_economy.pdf](#)

DST's Report

[dst_quantum_safe_ecosystem.pdf](#)

Society for Electronic Transactions and Security.

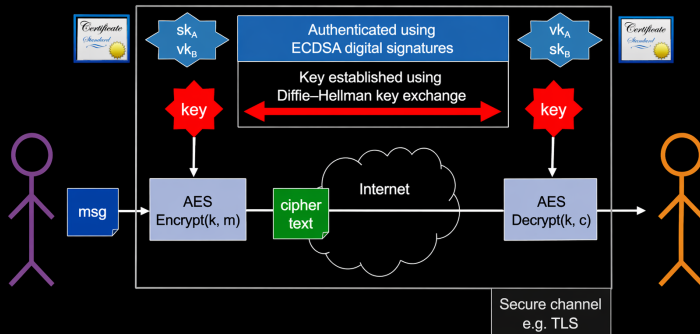
[pki_pqc_india.pdf](#)

CERT-In and SISA Whitepaper.

[certin_quantum_readiness.pdf](#)

[†] PDF documents are available in the ref directory

Authenticated key exchange + symmetric encryption



Don't worry if you don't understand all the terms!

Primitives and Constructions

- Primitives
 - Pseudorandom Generators (PRGs)
 - Hash functions
- Constructions
 - Key Derivation Functions (KDFs)
 - Message Authentication Codes (MACs)

Schemes and Protocols

– Schemes

- Digital signatures
- Symmetric-key encryption methods
- Public-key encryption

– Protocols

- Diffie-Hellman Key Exchange
- Transport Layer Security (TLS)
- Signal (*and* Whatsapp) Messaging Protocol

Cipher Suites

Signature Algorithms

Diffie-Hellman

Key Schedule

Questions?

Next Session: **Hands-on Classical Cryptography**